Bihar Engineering University, Patna

B.Tech 7th Semester Examination, 2024

Course: B.Tech
Code: 104702

Subject: Cost Management of Engineering Projects

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):-*** **[2 x 7 = 14]**

(a) The cost performance baseline is a time-phased budget and is used as a basis to measure, monitor and control overall cost performance of the project. It is usually displayed in form of
(i) An S- Curve
(ii) An inverted S- curve
(iii) Pie- chart
(iv) A Z-curve

(b) Contingency reserves are estimated costs to be used at the discretion of the project manager to deal with;
(i) Scope creep
(ii) Anticipated but not certain events
(iii) Unanticipated events
(iv) Anticipated and certain events

(c) A project is estimated to cost Rs 50,000 with a timeline of 50 days. After 25 days, the project manager finds that 50% of the project is complete an actual costs are Rs 50,000. What is the cost performance index (CPI)?
(i) The CPI is 1
(ii) The CPI is 1.5
(iii) The CPI is 2
(iv) The CPI is 0.5

(d) Your project is mid-way through a delivery schedule. As the project manager, you want to determine how much work is still left. Which is the most accurate way to determine the remaining work to be carried out by the project team?
(i) Rolling wave method
(ii) Earned value technique
(iii) A manual forecast
(iv) Future analysis

(e) An estimating technique that uses a statistical relationship between historical data and other variables is known as
(i) Parametric estimating
(ii) Analogous estimating
(iii) Bottom-up estimating
(iv) Historical analysis

(f) As a project manager, you periodically do project performance reviews to compare cost performance over time, schedule activities or work packages over-running an under-running budget, milestones due and milestones met. Which of the following is not a valid analytical technique?
(i) Tren Analysis
(ii) Earned value performance
(iii) Variance Analysis
(iv) Interpersonal skills

(g) A variance threshold for costs or other indicators to indicate the agreed amount of variation allowed is called
(i) Upper specification limit
(ii) Control thresholds
(iii) Upper control limits
(iv) Cost overrun

(h) In the earned value management technique, the cost performance baseline is referred to as;
(i) Cost Measurements Baseline
(ii) Performance Measurement Baseline
(iii) Performance Base Value
(iv) Performance cost baseline

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(i) As a part of EVM, a project manager is calculating the to-complete performance index (TCPI) based on EAC. The data that he has is: the budget at completion for the project is Rs 100,000. The earned value for the project is Rs 25000. The actual costs to date are Rs 40,000 and the estimate at completion is Rs 115,000. What is the TCPI that he will obtain?

(i) 1.1 (ii) 0.9
(iii) 1 (iv) 0

(j) A project uses rounding of activity estimates data to the nearest Rs 100, uses staff hours as the unit of measure for resources, and has a variance threshold of 10% deviation from the baseline plan. These would be typically documented in

(i) Cost management plan (ii) Scope Management
(iii) WBS (iv) Project Charter

Q.2 (a) Discuss the significance of bar charts and network diagram in project management. [7]
(b) Define the term "Project". Classify various types of projects. Discuss the need to manage projects. [7]

Q.3 (a) What is zero-based budgeting? State two advantages and two limitations each of this technique of budgeting. [7]
(b) Explain the Hungarian method for solving assignment problems. [7]

Q.4 (a) In the context of project network analysis, explain the similarities and differences between PERT and CPM. [7]
(b) State marginal costing with example. [7]

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Q.5 Given the following information; [14]
Fixed expenses = Rs 90,000/-
Variable cost per unit includes the three components as below;
Direct Material = Rs 5, Direct Labour = Rs 2, Direct Overheads = 100% of direct labour.
Selling price per unit = Rs 12.
Then calculate the followings;
(a) P/V Ratio ; (b) Breakeven sales in Rs with the help of P/V ratio; (c) Sales in Rs required to earn a profit of Rs 50,000/-.

Q.6 (a) What is Strategic Cost Management (SCM)? Explain the Stages of Strategic Cost Management [7]
(b) Distinguish between marginal costing and absorption costing? [7]

Q.7 (a) Explain Cost-Volume-Profit Analysis? What are different decision making problems? [7]
(b) Illustrate the major differences between Material Requirement Planning and Enterprise Resource Planning? [7]

Q.8 (a) How a project team is formed and discuss the role of each team member? [7]
(b) What is project execution as conglomeration of technical and non-technical activities? [7]

Q.9 Write Short notes on ***any two*** of the following:- [7 x 2 = 14]
(a) Classification of costs
(b) Objectives of costing system
(c) Performance budgets
(d) Budgetary control

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